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REDUCING DUTY CYCLE MISMATCH OF CLOCKS FOR CLOCK TRACKING CIRCUITS

Abstract

One or more examples relate to a method. The method may include: setting, at least partially based on an indication of mismatch between duty cycles of a first clock and a second clock generated to track the first clock, a duty cycle of a third clock; and providing the third clock having set duty cycle to an error detector of the clock tracking circuit in place of one of the first clock or the second clock.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/167,722, filed Feb. 10, 2023, now U.S. Pat. No. 12,308,845, issued May 20, 2025, which claims the benefit under 35 U.S.C. § 119 (e) of the priority date of U.S. Provisional Patent Application Ser. No. 63/367,434, filed Jun. 30, 2022, for DUTY CYCLE CORRECTION AND MATCHING TECHNIQUES, the contents and disclosure of each of which is incorporated herein in its entirety by this reference.

FIELD

[0002] One or more examples relate, generally, to clock tracking circuits for tracking an output clock to a reference clock. More specifically, one or more examples relate to reducing duty cycle mismatch between a reference clock and an output clock generated by a clock tracking circuit to track the reference clock.

BACKGROUND

[0003] Clock tracking circuits such as phase locked loops and delay locked loops, are circuits utilized to track clocks and other oscillating signals. An output signal of a clock tracking circuit is locked to the phase and frequency of a reference signal. Clock tracking circuits are utilized in a variety of operational contexts, including when two signals having known relationships are utilized to transmit information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

[0005] FIG. 1 is a block diagram depicting an apparatus to track a clock, in accordance with one or more examples.

[0006] FIG. 2 is a block diagram depicting an apparatus to change a duty cycle of a clock, in accordance with one or more examples.

[0007] FIG. 3 is a block diagram depicting an apparatus to generate a changed clock, and more specifically, a clock having a changed duty cycle compared to a duty cycle of an original clock, in accordance with one or more examples.

[0008] FIG. 4A is a timing diagram depicting signals during an example operation of the apparatus of FIG. 3 in accordance with one or more examples.

[0009] FIG. 4B is a timing diagram depicting signals during an example operation of the apparatus of FIG. 3 in accordance with one or more examples.

[0010] FIG. 5 is a diagram depicting a calibration process 500 in accordance with one or more examples.

[0011] FIG. 6 is a block diagram depicting an apparatus to indicate which of a first value, representing a duty cycle of a reference clock, or a second value, representing a duty cycle of an output clock (or feedback clock indicating the same), is greater, in accordance with one or more examples.

[0012] FIG. 7 is a block diagram depicting an apparatus to indicate which of a first value, representing a duty cycle of a reference clock, or a second value, representing a duty cycle of an

output clock (or feedback clock indicating the same), is greater, in accordance with one or more examples.

[0013] FIG. **8** is a flow diagram depicting a process to change a duty cycle of a clock to match the duty cycle of further clock, in accordance with one or more examples.

[0014] FIG. **9** is a flow diagram depicting a process to indicate which of a first value, representing a duty cycle of a reference clock, or a second value, representing a duty cycle of an output clock (or feedback clock indicating the same), is greater, in accordance with one or more examples.

[0015] FIG. **10** is a flow diagram depicting a process to set a duty cycle of a changed clock, in accordance with one or more examples.

[0016] FIG. **11** is a flow diagram depicting a process to generate a changed clock, and more specifically, a clock having a changed duty cycle compared to a duty cycle of an original clock, in accordance with one or more examples.

[0017] FIG. **12** is a block diagram depicting an apparatus to generate a changed clock, and more specifically, a clock having a changed duty cycle compared to a duty cycle of an original clock, in accordance with one or more examples.

[0018] FIG. **13A** and FIG. **13B** are timing diagrams respectively depicting signals during respective example operations of apparatus FIG. **12**, in accordance with one or more examples.

[0019] FIG. **14** is a flow diagram depicting a process to generate delay settings to set the amount of delay of a variable delay circuit or fixed delay circuits of FIG. **12**, in accordance with one or more examples.

[0020] FIG. **15** is a flow diagram depicting a process to generate a change clock via waveform shaping circuit of FIG. **12**, in accordance with one or more examples.

[0021] FIG. **16** is a block diagram of circuitry that, in some examples, may be used to implement various functions, operations, acts, processes, or methods disclosed herein.

DETAILED DESCRIPTION

[0022] In the following detailed description, reference is made to the accompanying drawings, which form a part hereof, and in which are shown, by way of illustration, specific examples of examples in which the present disclosure may be practiced. These examples are described in sufficient detail to enable a person of ordinary skill in the art to practice the present disclosure. However, other examples may be utilized, and structural, material, and process changes may be made without departing from the scope of the disclosure.

[0023] The illustrations presented herein are not meant to be actual views of any particular method, system, device, or structure, but are merely idealized representations that are employed to describe the examples of the present disclosure. The drawings presented herein are not necessarily drawn to scale. Similar structures or components in the various drawings may retain the same or similar numbering for the convenience of the reader; however, the similarity in numbering does not mean that the structures or components are necessarily identical in size, composition, configuration, or any other property.

[0024] The following description may include examples to help enable one of ordinary skill in the art to practice the disclosed examples. The use of the terms “exemplary,” “by example,” and “for example,” means that the related description is explanatory, and though the scope of the disclosure is intended to encompass the examples and legal equivalents, the use of such terms is not intended to limit the scope of an example of this disclosure to the specified components, steps, features, functions, or the like.

[0025] It will be readily understood that the components of the examples as generally described herein and illustrated in the drawing could be arranged and designed in a wide variety of different configurations. Thus, the following description of various examples is not intended to limit the scope of the present disclosure, but is merely representative of various examples. While the various aspects of the examples may be presented in drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

[0026] Furthermore, specific implementations shown and described are only examples and should not be construed as the only way to implement the present disclosure unless specified otherwise herein. Elements, circuits, and functions may be shown in block diagram form in order not to obscure the present disclosure in unnecessary detail. Conversely, specific implementations shown and described are exemplary only and should not be construed as the only way to implement the present disclosure unless specified otherwise herein. Additionally, block definitions and partitioning of logic between various blocks is exemplary of a specific implementation. It will be readily apparent to one of ordinary skill in the art that the present disclosure may be practiced by numerous other partitioning solutions. For the most part, details concerning timing considerations and the like have been omitted where such details are not necessary to obtain a complete understanding of the present disclosure and are within the abilities of persons of ordinary skill in the relevant art.

[0027] Those of ordinary skill in the art would understand that information and signals may be represented using any of a variety of different technologies and techniques. Some drawings may illustrate signals as a single signal for clarity of presentation and description. It will be understood by a person of ordinary skill in the art that the signal may represent a bus of signals, wherein the bus may have a variety of bit widths and the present disclosure may be implemented on any number of data signals including a single data signal.

[0028] The various illustrative logical blocks, modules, and circuits described in connection with the examples disclosed herein may be implemented or performed with a general purpose processor, a special purpose processor, a Digital Signal Processor (DSP), an Integrated Circuit (IC), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor (may also be referred to herein as a host processor or simply a host) may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. A general-purpose computer including a processor is considered a special-purpose computer while the general-purpose computer is configured to execute computing instructions (e.g., software code) related to examples of the present disclosure.

[0029] The examples may be described in terms of a process that is depicted as a flowchart, a flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe operational acts as a sequential process, many of these acts can be performed in another sequence, in parallel, or substantially concurrently. In addition, the order of the acts may be re-arranged. A process may correspond to a method, a thread, a function, a procedure, a subroutine, a subprogram, without limitation. Furthermore, the methods disclosed herein may be implemented in hardware, software, or both. If implemented in software, the functions may be stored or transmitted as one or more instructions or code on computer-readable media. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another.

[0030] Any reference to an element herein using a designation such as “first,” “second,” and so forth does not limit the quantity or order of those elements, unless such limitation is explicitly stated. Rather, these designations may be used herein as a convenient method of distinguishing between two or more elements or instances of an element. Thus, a reference to first and second elements does not mean that only two elements may be employed there or that the first element must precede the second element in some manner. In addition, unless stated otherwise, a set of elements may comprise one or more elements.

[0031] As used herein, the term “substantially” in reference to a given parameter, property, or

condition means and includes to a degree that one of ordinary skill in the art would understand that the given parameter, property, or condition is met with a small degree of variance, such as, for example, within acceptable manufacturing tolerances. By way of example, depending on the particular parameter, property, or condition that is substantially met, the parameter, property, or condition may be at least 90% met, at least 95% met, or even at least 99% met.

[0032] As used herein, any relational term, such as “over,” “under,” “on,” “underlying,” “upper,” “lower,” without limitation, is used for clarity and convenience in understanding the disclosure and accompanying drawings and does not connote or depend on any specific preference, orientation, or order, except where the context clearly indicates otherwise.

[0033] In this description the term “coupled” and derivatives thereof may be used to indicate that two elements co-operate or interact with each other. When an element is described as being “coupled” to another element, then the elements may be in direct physical or electrical contact or there may be intervening elements or layers present. In contrast, when an element is described as being “directly coupled” to another element, then there are no intervening elements or layers present. The term “connected” may be used in this description interchangeably with the term “coupled,” and has the same meaning unless expressly indicated otherwise or the context would indicate otherwise to a person having ordinary skill in the art.

[0034] A clock signal or just “clock,” is a signal that oscillates between a high state and a low state in a reliably predictable manner. A circuit can be responsive to a rising or falling edge of a clock to coordinate its actions.

[0035] Phase locked loops (PLLs) are used, generally, to generate an output clock that tracks a reference clock—e.g., are phase and frequency locked to the reference clock signal, without limitation. In a typical clock tracking circuit, an oscillator is controlled via one or more control paths to generate an output clock having the desired phase and frequency. When the frequency is different from the desired frequency (also called “frequency offset” or “frequency error”), the clock tracking circuit generates control signals via the one or more control paths to urge the oscillator's native frequency toward the desired frequency. At certain frequency ranges, a clock tracking circuit may exhibit jitter, which is an integration of phase noise in specific bandwidths.

[0036] Duty cycle distortion and duty cycle mismatch may contribute to jitter or spurs in a clock tracking circuit that compares an output clock (or a feedback signal representative of an output clock signal) to a reference clock. Duty cycle distortion is a difference between the duty of a clock signal and an ideal duty cycle, which ideal duty cycle may be 50%. Duty cycle mismatch is a difference between respective duty cycles of oscillating signals (e.g., clock signals) that are compared at least in terms of phase or frequency. Some clock tracking circuits individually correct the duty cycle distortion present in the reference clock, output clock, or both, and assume it will correct duty cycle mismatch between a reference clock and an output clock. However, the inventors of this disclosure appreciate that when the results of correcting duty-cycle distortion individually in two clock signals are combined, e.g., that of the reference clock and that of the output clock, errors incurred during the respective duty-cycle distortion correction processes is added—increasing overall error.

[0037] The inventors of this disclosure appreciate that duty cycle mismatch correction that utilizes both clock signals would be desirable, as a non-limiting example, by avoiding or reducing at least some of undesirable effects of applying separate processes to correct duty-cycle distortion, discussed above.

[0038] FIG. 1 is a block diagram depicting an apparatus **100** to track a clock (and may also be referred to herein as a “clock tracking circuit **100**”), in accordance with one or more examples. As non-limiting examples, clock tracking circuit **100** may be a hybrid phase locked loop (PLL), an analog PLL, a digital PLL, a delay locked loop (DLL), an injection locked loop (ILL), or a frequency synthesizer.

[0039] Apparatus **100** includes an error detector **102**, a controller **104**, a controlled-oscillator **106**,

and a duty cycle matching circuit **108**. Apparatus **100** operates, generally, to generate an output clock **114** phase-locked and frequency-locked to a reference clock **110**.

[0040] Error detector **102** generates an error signal **118** that includes information about a phase and frequency difference between feedback clock **116** and reference clock **110**. As non-limiting examples, error detector **102** may be or include one or more of a phase-frequency detector, a bang-bang phase detector, a time-to-digital converter, or a subsampling phase detector. Reference clock **110** may be generated by any suitable clock source for a given operational context. Feedback clock **116** may be the same as output clock **114** (e.g., output clock **114** is fed directly to an input of duty cycle matching circuit **108**, without limitation) or a signal indicative of the phase, frequency, pulse width, and duty cycle of output clock **114**. For example, the phase, frequency, pulse width, and duty cycle of feedback clock **116** may be the same or different than output clock **114**, but in either case, is relatable back to the phase, frequency, pulse width, and duty cycle of output clock **114**. As a non-limiting example, feedback clock **116** may be a frequency divided version of output clock **114** (frequency divider/buffer not depicted).

[0041] Controlled-oscillator **106** is an electronic oscillator for generating output clock **114** at least partially in response to control signal **120**, which may be digital control signals (i.e., in the case of a digital controlled oscillator (DCO)), voltage control signals (i.e., in the case of a voltage controlled oscillator (VCO)), current control signals (i.e., in the case of a current controlled oscillator (CCO)) or a combination thereof (i.e., a hybrid controlled oscillator (e.g., any combination of digital, voltage, or current controlled, without limitation)).

[0042] In one or more examples, error signal **118** includes direction information and magnitude information. Direction information of error signal **118** indicates if output clock **114** lags in phase or leads in phase with respect to reference clock **110**. Magnitude information of error signal **118** indicates an extent to which output clock **114** lags in phase or leads in phase with respect to reference clock **110** (i.e., an amount of phase difference). In one or more examples, error signal **118** may be utilized to determine phase and frequency information about output clock **114** and reference clock **110**, and information about the differences there between.

[0043] In the specific non-limiting example depicted by FIG. **1**, apparatus **100** includes a controller **104** to provide control signal **120**, at least partially in response to error signal **118**, to controlled-oscillator **106** to adjust output clock **114**. Controller **104** may include circuits (analog circuits, digital circuits, or both) to provide proportional control and integral control of controlled-oscillator **106**. Controller **104** may include a proportional control path for transient correction of phase differences between feedback clock **116** and reference clock **110** indicated by error signal **118**. Controller **104** may include an integral control path to urge an average frequency of controlled-oscillator **106** toward a target frequency (e.g., a frequency of reference clock **110** or a multiple thereof, without limitation) in response to frequency differences between feedback clock **116** and reference clock **110** indicated by error signal **118**.

[0044] Duty cycle matching circuit **108** provides matched clocks **112** including versions of reference clock **110** and feedback clock **116** having matched duty cycles as discussed herein. In one or more examples, duty cycle matching circuit **108** may change respective duty cycles of one or both of reference clock **110** and feedback clock **116**. Matched clocks **112** may exhibit reduced duty cycle mismatch than exhibited between feedback clock **116** and reference clock **110**. Reduced duty cycle mismatch may reduce jitter or spurs exhibited by apparatus **100**, e.g., by output clock **114**.

[0045] FIG. **2** is a block diagram depicting an apparatus **200** to change a duty cycle of a clock, in accordance with one or more examples. Apparatus **200** is a non-limiting example of a duty cycle matching circuit **108** of FIG. **1**, and may also be referred to as a “duty cycle matching circuit **200**.”

[0046] Apparatus **200** generates a changed clock **212** having a duty cycle set to reduce duty cycle mismatch between the changed clock **212** and a target clock **204**. Changed clock **212** may be provided in lieu of original clock **214**. The duty cycle mismatch between changed clock **212** and target clock **204** may be less than the duty cycle mismatch between original clock **214** and target

clock **204**. In this manner, when changed clock **212** is provided in lieu of original clock **214**, apparatus **200** may be understood to reduce duty cycle mismatch between original clock **214** and target clock **204**. In one or more examples, “changing a duty cycle of a clock” includes, but is not limited to, generating a clock having a set duty cycle.

[0047] Changing a duty cycle of a clock may also be referred to herein as “calibrating a duty cycle of a clock” and process to do the same referred to herein as a “calibration process.” In one or more examples, a calibration process may include multiple calibration cycles, during at least some of which a duty cycle of the changed clock **212** is incrementally set, as discussed below. The number of calibration cycles implemented is a matter of design choice and may be set, as a non-limiting example, based on specific operating conditions. In one or more examples, a number of calibration cycles may be variable, as a non-limiting example, when a given calibration process executes until duty cycle mismatch is below a predetermined threshold. In one or more examples, a number of calibration cycles may be variable up to a predetermined upper limit. In one or more examples, a number of calibration cycles may be variable with a predetermined lower limit (e.g., at least the lower limit number of calibration cycles are executed before determining duty cycle mismatch is below the predetermined threshold, without limitation).

[0048] In one or more examples, logic circuit **208** and duty cycle matching circuit **206** may be synchronized via local clock **220**. Further, calibration cycles may be synchronized to clock cycles of local clock **220**.

[0049] Apparatus **200** includes a comparison circuit **202**, a logic circuit **208**, and a duty cycle matching circuit **206** arranged in a signal path **210** of original clock **214**. Original clock **214** is the clock to be changed. In one or more examples, original clock **214** may be feedback clock **116** or reference clock **110** of FIG. 1, and target clock **204** may be the other of feedback clock **116** or reference clock **110** of FIG. 1.

[0050] Comparison circuit **202** receives changed clock **212** and target clock **204**, and generates mismatch signal **216** to indicate which of the respective duty cycles of changed clock **212** and target clock **204** is greater. Changed clock **212** is the output of apparatus **200**. More specifically, mismatch signal **216** may indicate that respective duty cycles of changed clock **212** and target clock **204** are equal or not equal. If not equal, mismatch signal **216** may indicate which of the duty cycle of changed clock **212** or the duty cycle of target clock **204** is greater. A greater duty cycle may also be characterized herein as a “wider” duty cycle, since a pulse width of the clock having the greater duty cycle will be wider. Indicating the greater duty cycle implies the other duty cycle is lesser and the respective pulse width “narrower.”

[0051] Logic circuit **208** generates matching settings **218** to set internal circuitry of duty cycle matching circuit **206** in a manner that increases or decreases a duty cycle of changed clock **212**. In one or more examples, logic circuit **208** may determine matching settings **218** to increase or decrease a duty cycle of changed clock **212** by a predetermined amount over one or more calibration cycles of a calibration process. Over multiple calibration cycles, a duty cycle of changed clock **212** changes toward a duty cycle of target clock **204**. If mismatch signal **216** indicates a duty cycle of changed clock **212** is greater than a duty cycle of target clock **204**, logic circuit **208** generates matching settings **218** to decrease a duty cycle of changed clock **212** by the predetermined amount. If mismatch signal **216** indicates a duty cycle of changed clock **212** is lesser than a duty cycle of logic circuit **208**, logic circuit **208** generates matching settings **218** to increase a duty cycle of changed clock **212** by the predetermined amount.

[0052] Duty cycle matching circuit **206** is arranged in a signal path **210** of original clock **214** and generates changed clock **212** at least partially responsive to original clock **214** and the matching settings **218** generated by logic circuit **208**. By the end of a given calibration process, changed clock **212** is a duty cycle matched version of original clock **214**. In various examples, a calibration process may execute until a duty cycle of changed clock **212** matches a duty cycle of target clock **204**. Alternatively, in one or more examples, a calibration process operation may execute for a

predetermined number of calibration cycles and then end. While the calibration process executes, a duty cycle of changed clock **212** may change to be generally more like a duty cycle of target clock **204**, though respective duty cycles may not necessarily match at the end of the calibration process. Stated another way, while the calibration process executes, a mismatch between respective duty cycles of changed clock **212** and target clock **204** may decrease, though not be completely eliminated.

[0053] In one or more examples, changed clock **212** may be provided in lieu of reference clock **110** or output clock **114** of FIG. **1**, as the case may be.

[0054] FIG. **3** is a block diagram depicting an apparatus **300** to generate a changed clock, and more specifically, a clock having a changed duty cycle compared to a duty cycle of an original clock, and more specifically still, a clock having a changed pulse width compared to a pulse width of the original clock, in accordance with one or more examples.

[0055] Apparatus **300** includes a variable delay circuit **302**, a fixed delay circuit **304**, a comparison circuit **306**, a logic circuit **308**, and a waveform shaping circuit **338**. Waveform shaping circuit **338** includes a wide logic circuit **320**, a skinny logic circuit **322**, and a multiplexer **310**. Comparison circuit **306** is similar to comparison circuit **202** of FIG. **2** and is not re-described, here. Further, changed clock **318**, target clock **340**, and original clock **312** are similar to changed clock **212**, target clock **204** and original clock **214** of FIG. **2**, and are not re-described, here.

[0056] Variable delay circuit **302** receives delay setting **330** and original clock **312**, and generates first delayed clock **324** at least partially responsive to delay setting **330** and original clock **312**. A respective delay implemented by variable delay circuit **302** corresponds to delay setting **330** generated by logic circuit **308**.

[0057] Fixed delay circuit **304** receives original clock **312** and generates second delayed clock **326** at least partially responsive to original clock **312**. Fixed delay circuit **304** implements a fixed delay. In one or more examples, a delay implemented by fixed delay circuit **304** may be hard wired or coded, or fixed delay circuit **304** may be a variable delay circuit set by a control signal (e.g., generated by logic circuit **308** or a further logic circuit that is not depicted, without limitation) that indicates a substantially constant delay. In one or more examples, a delay implemented by fixed delay circuit **304** corresponds to an internal delay of apparatus **300**.

[0058] In an initial state, variable delay circuit **302** implements substantially the same fixed delay implemented by fixed delay circuit **304**.

[0059] In one or more examples, variable delay circuit **302** and fixed delay circuit **304** may internally implement any suitable technique to generate first delayed clock **324** and second delayed clock **326**, respectively, based on original clock **312**. A non-limiting example of a suitable technique may include enabling a number of buffers or inverters in a signal path to increase delay and disabling a number of buffers or inverters in the signal path to decrease delay.

[0060] In one or more examples, variable delay circuit **302** or fixed delay circuit **304** may respectively include delay cells **334**/delay cells **336** to implement respective delays. In one or more examples, delay cells **334**/delay cells **336** may utilize any suitable technique to implement delay. By way of non-limiting example, delay cells **334**/delay cells **336** may be or include logic gates or devices (e.g., inverters, buffers, without limitation) that enable cascaded connection of a variable number of such gates or devices. By way of another non-limiting example, delay cells **334**/delay cells **336** may be or include circuits, such as resistor-capacitor circuits where a variable number of resistors are selectively coupled in parallel. Delay exhibited by a resistor-capacitor circuit is a function of the time constant of the circuit, and is inversely proportional to the number of resistors selectively coupled in parallel.

[0061] First delayed clock **324** and second delayed clock **326** are both provided to waveform shaping circuit **338**, and wide logic circuit **320** and skinny logic circuit **322**, more specifically. Generally, waveform shaping circuit **338** performs waveform shaping as discussed below, e.g., waveform shaping circuit **338** generate changed clock **318** with a changed pulse width and,

accordingly, a changed duty cycle.

[0062] Wide logic circuit **320** and skinny logic circuit **322** respectively provide waveform shaping functions of waveform shaping circuit **338**. Wide logic circuit **320** shortens wide pulses (e.g., generates first changed clock **314** with a waveform having a shorter pulse width, and therefore lower duty cycle than first delayed clock **324**, which first delayed clock **324** has the same pulse width as original clock **312**), and skinny logic circuit **322** widens short pulses (e.g., generates second changed clock **316** with a waveform having a wider pulse width, and therefore greater duty cycle, than second delayed clock **326**, which second delayed clock **326** has the same pulse width as original clock **312**), as discussed below. Here, “wide” refers to a pulse width of changed clock **318** being longer than a desired pulse width, and “skinny” refers to a pulse width of changed clock **318** being shorter than a desired pulse width. As used herein, a “desired duty cycle” is a duty cycle of target clock **340** and a “desired pulse width” is a pulse width of target clock **340**.

[0063] Multiplexer **310** selects one of first changed clock **314** or second changed clock **316** at least partially responsive to waveform shaping setting **332** generated by logic circuit **308**. In this manner, waveform shaping setting **332** sets a waveform shaping function of waveform shaping circuit **338**.

[0064] Logic circuit **308** generates delay setting **330** and waveform shaping setting **332** to reduce duty cycle mismatch between changed clock **318** and target clock **340** as indicated by mismatch signal **328**. In one or more examples, matching settings **218** of FIG. 2 include delay setting **330** and waveform shaping setting **332**.

[0065] The first delayed clock **324** differs in phase from second delayed clock **326** by an amount that corresponds to a delay implemented at variable delay circuit **302** in response to delay setting **330**. Waveform shaping circuit **338**, and more specifically wide logic circuit **320** and skinny logic circuit **322**, as the case may be, utilize the amount of phase difference to set a pulse width of changed clock **318**, and accordingly, set a duty cycle of changed clock **318**.

[0066] In one or more examples, a delay amount by which variable delay circuit **302** is incremented or decremented may be fixed per calibration cycle to a predetermined amount (also referred to herein as a “delay step size”). For a given calibration cycle, the predetermined delay amount may be less than a total difference between a pulse width of changed clock **318** and a desired pulse width (or a pulse width of original clock **214** and target clock **204**). Apparatus **300** may execute multiple calibration cycles to generate a changed clock **318** having desired pulse width or a changed clock **318** having a pulse width exhibiting reduced duty cycle mismatch to the duty cycle of target clock **340**. Operation of logic circuit **308** and variable delay circuit **302** may be synchronized by local clock **342**, and respective calibration cycles may correspond to clock cycles of local clock **342**.

[0067] In various examples, a total number of calibration cycles may be limited to a predetermined number (a predetermined upper limit) and, in some cases, a pulse width of changed clock **318** that results from a calibration process may, or may not, substantially match a pulse width of target clock **340**.

[0068] For a given calibration cycle, if mismatch signal **328** indicates a duty cycle of changed clock **318** is greater than a duty cycle of target clock **340**, logic circuit **308** generates delay setting **330** and waveform shaping setting **332** to shorten a pulse width of changed clock **318** by a predetermined amount. If mismatch signal **328** indicates a duty cycle of changed clock **318** is less than a duty cycle of target clock **340**, logic circuit **308** generates delay setting **330** and waveform shaping setting **332** to lengthen a pulse width of changed clock **318** by a predetermined amount.

[0069] If logic circuit **308** determines that a pulse width of second delayed clock **326** is skinnier than a desired pulse width, logic circuit **308** generates waveform shaping setting **332** to select second changed clock **316** input at multiplexer **310**. If logic circuit **308** determines a pulse width of second delayed clock **326** is wider than a desired pulse width, logic circuit **308** generates waveform shaping setting **332** to select first changed clock **314** input at multiplexer **310**.

[0070] FIG. 4A and FIG. 4B are timing diagrams depicting signals **400** during respective example

operations of apparatus **300**. Specifically, FIG. 4A depicts second delayed clock **326**, first delayed clock **324**, and first changed clock **314** in an example operation of wide logic circuit **320**, and FIG. 4B depicts second delayed clock **326**, first delayed clock **324**, and second changed clock **316** in an example operation of skinny logic circuit **322**. In these examples, wide logic circuit **320** and skinny logic circuit **322** operate as, or are, an AND-gate and an OR-gate, respectively.

[0071] Turning to an example shortening of a wide pulse by wide logic circuit **320** depicted by FIG. 4A:

[0072] At time T.sub.1, second delayed clock **326** exhibits a change from a first state to a second, different, state, here a rising edge, while first delayed clock **324** remains in a first state. At time T.sub.1, wide logic circuit **320** generates first changed clock **314** in a first state because at least one of second delayed clock **326** and first delayed clock **324** are in the first state.

[0073] At time T.sub.2, first delayed clock **324** exhibits a rising edge. A time duration ΔT from time T.sub.1 to time T.sub.2 is substantially equal to one or more delay increments, discussed above. Time duration ΔT may correspond to one or more delay increments of duty cycle matching circuit **206** or variable delay circuit **302**.

[0074] Due to delay setting **330**, a rising edge of first delayed clock **324** is “behind” (in terms of time) a rising edge of second delayed clock **326**. At time T.sub.2, wide logic circuit **320** generates first changed clock **314** in a second state because second delayed clock **326** and first delayed clock **324** exhibit the same, second, state.

[0075] At time T.sub.3, second delayed clock **326** exhibits a change from a second state to a first state, here a falling edge, while first delayed clock **324** remains in a second state. At time T.sub.3, wide logic circuit **320** generates a falling edge for first changed clock **314**, i.e., first changed clock **314** changes to the first state, because at least one of second delayed clock **326** and first delayed clock **324** are in the first state. The time duration between times T.sub.2 and T.sub.3, i.e., the pulse width of first changed clock **314**, is shorter than the time duration between T.sub.1 and T.sub.3, i.e., the pulse width of second delayed clock **326**.

[0076] Turning to an example widening of a short pulse by skinny logic circuit **322** depicted by FIG. 4B:

[0077] At time T.sub.1, second delayed clock **326** exhibits a change a change from a first state to a second, different, state, here a rising edge, while first delayed clock **324** remains in a first state. At time T.sub.1, skinny logic **322** generates second changed clock **316** exhibiting a change from a first state to a second state because at least one of second delayed clock **326** and first delayed clock **324** are in the second state.

[0078] At time T.sub.2, second delayed clock **326** exhibits a change from the first state to the second state (a rising edge), while first delayed clock **324** remains in the second state, so skinny logic circuit **322** continues to generate second changed clock **316** exhibiting the second state. Notably, the time duration between time T.sub.1 and time T.sub.2 may substantially equal one or more delay increments implemented by variable delay circuit **302**, discussed above.

[0079] At time T.sub.3, second delayed clock **326** exhibits a change from the second state to the first state (falling edge), while first delayed clock **324** remains in the second state, so skinny logic circuit **322** continues to generate second changed clock **316** exhibiting the second state because at least one of second delayed clock **326** and first delayed clock **324** are in the second state.

[0080] At time T.sub.4, first delayed clock **324** exhibits a falling edge from the second state to the first state, while second delayed clock **326** remains in the first state, so skinny logic **322** generates second changed clock **316** exhibiting a falling edge from the second state to the first state because both second delayed clock **326** and first delayed clock **324** are in the first state. The time duration between times T.sub.1 and T.sub.4 is longer than the pulse width of either second delayed clock **326** or second delayed clock **326**.

[0081] FIG. 5 is a diagram depicting a calibration process **500** in accordance with one or more examples. Depicted are delay line update numbers (“Delay Line update #”) which identify a

sequence of respective calibration cycles of a calibration process **500**. Also depicted are respective delay values (“Delay Line Value”) applied during the respective calibration cycles. Each delay line value represents a multiple of a predetermined delay amount. For example, “1” represents 1×the predetermined delay amount, “2” represents 2×the predetermined delay amount, and so on and so forth.

[0082] In the specific example depicted by FIG. 5, respective delay values increment from delay line update number 0 to delay line update number 5, decrement from delay line update number 5 to delay line update number 7, and increment from delay line update number 7 to delay line update number 8. From delay line update number 8 to 1023, the delay line update number toggles between incrementing and decrementing, which indicates duty cycles are matched for purposes of calibration process **500**.

[0083] FIG. 6 is a block diagram depicting an apparatus **600** to indicate which of a first value, representing a duty cycle of a reference clock, or a second value, representing a duty cycle of an output clock (or feedback clock indicating the same), is greater, in accordance with one or more examples. Apparatus **600** is a non-limiting example of a comparison circuit **202** of FIG. 2 and comparison circuit **306** of FIG. 3, and may be referred to as a “comparison circuit **600**.”

[0084] Apparatus **600** includes first filter **602**, second filter **604**, and comparator circuit **606**. First filter **602** filters changed clock **608** to obtain a first DC signal **612**, which is an analogue signal at a signal level proportional to a pulse width of changed clock **608** in relation to a total clock cycle of changed clock **608**, and thus, represents a duty cycle of changed clock **608**. Second filter **604** filters target clock **610** to obtain second DC signal **614**, which is an analogue signal at a signal level proportional to a pulse width of target clock **610** in relation to a total clock cycle of target clock **610**, and thus represents a duty cycle of target clock **610**.

[0085] In one or more examples, first filter **602** and second filter **604** may be resistor-capacitor (RC) low-pass filter circuits. The RC circuits convert the oscillating or pulsed changed clocks **608** and **610** to a substantially fixed voltage level signal (e.g., a direct current (DC) signal). As a non-limiting example first filter **602** or second filter **604** may generate a DC voltage signal between 0 volts and 1 volt, inclusive (or a voltage level between 0 volts and a voltage level of a supply voltage). The specific DC voltage level generated by the first and second filters **602**, **604** may be proportional to a duty cycle of changed clock **608** or target clock **610**, respectively.

[0086] Comparator circuit **606** generates an output (mismatch signal **616**) exhibiting a first state if first DC signal **612** is greater than second DC signal **614**, a second state if second DC signal **614** is greater than first DC signal **612**, and a third state if first DC signal **612** and second DC signal **614** are the same. In one or more examples, the third state is a toggling signal that changes between the first and second state due to non-ideal characteristics of comparison comparator circuit **606**.

[0087] FIG. 7 is a block diagram depicting apparatus **700** to indicate which of a first value, representing a duty cycle of a reference clock, or a second value, representing a duty cycle of an output clock (or feedback clock indicating the same), is greater, in accordance with one or more examples, as well as an amount of the difference between the duty cycle of the reference clock and the duty cycle of the output clock, as discussed above.

[0088] Apparatus **700** includes comparison circuit **720** and duty cycle matching circuit **718**. Comparison circuit **720** includes first filter **702** and second filter **704**, which are similar to first filter **602** and second filter **604** and aren't re-described here; and differential amplifier **706**. Differential amplifier **706** generates a mismatch signal **716** that is proportional to a difference between first DC signal **712** and second DC signal **714**. First DC signal **712** is representative of a duty cycle of changed clock **212** and second DC signal **714** is representative of a duty cycle of target clock **710**.

[0089] Mismatch signal **716** is provided directly to a control input **724** of duty cycle matching circuit **718** (e.g., an output of differential amplifier **706** is directly coupled to an input of duty cycle matching circuit **718**, without limitation), and more specifically to a variable delay circuit of duty

cycle matching circuit **718**, which may be of a waveform shaping circuit of duty cycle matching circuit **718** (respectively not depicted). In this manner, duty cycle matching circuit **718** may generate changed clock **708** at least partially responsive to original clock **722** and the output of differential amplifier **706** (mismatch signal **716**).

[0090] FIG. **8** is a flow diagram depicting a process **800** to change a duty cycle of a clock to match the duty cycle of a further clock, in accordance with one or more examples. As a non-limiting example, some or a totality of process **800** may be performed by duty cycle matching circuit **108**, apparatus **200** or apparatus **300**.

[0091] Although FIG. **8** depicts process **800** as a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. In one or more examples, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of process **800**. In one or more examples, different components of an apparatus or system implementing some or a totality of operations of process **800** may perform functions at substantially the same time or in a specific sequence.

[0092] At operation **802**, process **800** compares (e.g., via a comparison circuit **202**, without limitation) a first value and a second value. The first value represents a duty cycle of a reference clock, and the second value represents a duty cycle of an output clock generated by a clock tracking circuit to track the reference clock.

[0093] At **804**, process **800** sets (e.g., via matching settings **218** provided to duty cycle matching circuit **206** or via an output of comparison circuit **720** provided directly to duty cycle matching circuit **718**, without limitation) a duty cycle of a changed clock (e.g., changed clock **212**, without limitation) to reduce duty cycle mismatch between the reference clock and the output clock as indicated by the comparison of the first value and the second value. In one or more examples, process **800** may set a duty cycle of the changed clock via a calibration process that includes multiple calibration cycles. Duty cycle mismatch may be incrementally reduced at least partially responsive to respective calibration cycles.

[0094] At operation **806**, process **800** provides the changed clock (e.g., provides changed clock **212**, without limitation) having the set duty cycle in lieu of the one of the reference clock or the output clock (e.g., as a replacement for reference clock **110** or feedback clock **116** as respective inputs of error detector **102**, without limitation).

[0095] FIG. **9** is a flow diagram depicting a process **900** to generate values representing respective pulse widths, and thus, respective duty cycles of the reference clock and the output clock generated the clock tracking circuit, in accordance with one or more examples. As a non-limiting example, some or a totality of operations of process **900** may be performed by apparatus **600**.

[0096] Although FIG. **9** depicts process **900** as a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. In one or more examples, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of process **900**. In one or more examples, different components of an apparatus or system implementing some or a totality of process **900** may perform functions at substantially the same time or in a specific sequence.

[0097] At operation **902**, process **900** filters (e.g., via first filter **602**, without limitation) a reference clock to obtain a first DC signal (e.g., first DC signal **612**, without limitation). The first DC signal generated by filtering the reference clock may have a value proportional to a pulse width of the reference clock, and accordingly, a duty cycle of the reference clock in relation to a clock cycle of the reference clock.

[0098] At operation **904**, process **900** filters (e.g., via second filter **604**, without limitation) the output clock to obtain a second DC signal (e.g., second DC signal **614**, without limitation). The second DC signal generated by filtering the output clock may have a value proportional to a pulse width of the output clock, and accordingly, a duty cycle of the output clock in relation to a clock cycle of the output clock.

[0099] At optional operation **906** of process **900**, the first value of operation **902** corresponds to the first DC signal and the second value of operation **902** corresponds to the second DC signal.

[0100] FIG. **10** is a flow diagram depicting a process **1000** to set a duty cycle of a changed clock, in accordance with one or more examples.

[0101] FIG. **10** illustrates an example process to set a duty cycle of a changed clock, in accordance with one or more examples. Although the example routine depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of process **1000**. In other examples, different components of an example device or system that implements the routine may perform functions at substantially the same time or in a specific sequence.

[0102] At operation **1002**, process **1000** increments or decrements (e.g., via delay setting **330** generated by logic circuit **308**, without limitation) a delay at a variable delay circuit (e.g., variable delay circuit **302**, without limitation) at least partially responsive to a comparison of a first value and a second value. The first value represents a duty cycle of a reference clock and the second value represents a duty cycle of an output clock generated by a clock tracking circuit to track the reference clock. The amount of delay by which the variable delay circuit is incremented or decremented may be a predetermined amount of delay, i.e., a delay step size.

[0103] At operation **1004**, process **1000** delays one of the reference clock or the output clock at the variable delay circuit to generate a first delayed clock (e.g., first delayed clock **324**, without limitation). The delay implemented by the variable delay circuit to delay the reference clock or output clock and generate the first delayed clock **324** has been increased or decreased by the amount of delay by which the variable delay circuit was incremented or decremented in operation **1002**.

[0104] At operation **1006**, process **1000** delays the one of the reference clock or the output clock at a fixed delay circuit (e.g., fixed delay circuit **304**, without limitation) to generate a second delayed clock (e.g., second delayed clock **326**, without limitation).

[0105] At operation **1008**, process **1000** shapes a waveform of a changed clock at least partially responsive to the first delayed clock and the second delayed clock. Waveform shaping may be performed at least partially responsive to the comparison of the first value and the second value and a determination of whether to shorten or lengthen a pulse width, and accordingly a duty cycle, of the changed clock.

[0106] FIG. **11** is a flow diagram depicting a process **1100** to generate a changed clock, and more specifically, a changed clock having a shaped waveform exhibiting a duty cycle that is different than a duty cycle of an original clock, in accordance with one or more examples. Some or a totality of operations of process **1100** may be performed by logic circuit **208** or logic circuit **308**.

[0107] Although the example routine depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of process **1000**. In other examples, different components of an example device or system that implements the routine may perform functions at substantially the same time or in a specific sequence.

[0108] At operation **1102**, process **1100** generates a delay setting for a variable delay circuit arranged in a portion of the signal path of one of the reference clock or the output clock. The delay indicated by the delay setting may represent a predetermined amount of delay, i.e., a predetermined multiple of a delay step size.

[0109] At operation **1104**, if the desired pulse width is longer than the pulse width of the original clock, process **1100** generates a waveform shaping setting indicating the changed clock having a lengthened pulse width.

[0110] At operation **1106**, if the desired pulse width is shorter than the pulse width of the original

clock, process **1100** generates a waveform shaping setting indicating the changed clock having a shortened pulse width.

[0111] At optional operation **1108**, optionally provide the generated waveform shaping setting (e.g., waveform shaping setting **332**, without limitation) to a waveform shaping circuit (e.g., waveform shaping circuit **338**, without limitation).

[0112] FIG. **12** is a block diagram depicting an apparatus **1200** to generate a changed clock, and more specifically, a clock having a changed duty cycle compared to a duty cycle of an original clock, and more specifically still, a clock having a changed pulse width compared to a pulse width of the original clock, in accordance with one or more examples. Apparatus **300** is a non-limiting example of apparatus **200** or duty cycle matching circuit **108**.

[0113] Apparatus **1200** includes a variable delay circuit **1202**, a fixed delay circuit **1204**, a comparison circuit **1206**, a logic circuit **1208**, and a waveform shaping circuit **1222**. Comparison circuit **1206** is similar to comparison circuit **202** of FIG. **2** and is not re-described, here. Further, variable delay circuit **1202**, delay cells **1218**, fixed delay circuit **1204**, and delay cells **1220** are similar to variable delay circuit **302**, delay cells **334**, fixed delay circuit **304**, and delay cells **336** are not re-described here. Further, local clock **1226**, changed clock **1212**, target clock **1224**, and original clock **1210** are similar to local clock **342**, changed clock **212**, target clock **204** and original clock **214** of FIG. **2**, and are not re-described, here.

[0114] Initially, respective delays at variable delay circuit **1202** and fixed delay circuit **1204** may be set to $\frac{1}{2}$ the delay range of variable delay circuit **1202**. Once set at $\frac{1}{2}$ the delay range, the delay amount at variable delay circuit **1202** may be incremented or decremented by one or more delay amounts by logic circuit **1208** via delay setting **1216**.

[0115] Waveform shaping circuit **1222** generates changed clock **1228** at least partially responsive to signals received at a first input **1234** and a second input **1236** of waveform shaping circuit **1222**. More specifically, waveform shaping circuit **1222** generates changed clock **1228** exhibiting a change from a first state to a second state (a rising edge) at least partially responsive to a signal received at first input **1234**, and generates changed clock **1228** exhibiting a change from a second state to a first state (a falling edge) at least partially responsive to a signal received at second input **1236**. First input **1234** may be characterized as the “rising edge control input” of waveform shaping circuit **1222** and second input **1236** may be characterized as the “falling edge control input” of waveform shaping circuit **1222**.

[0116] First input **1234** is coupled to an output of variable delay circuit **1202** and second input **1236** is coupled to an output of fixed delay circuit **1204**, accordingly, first input **1234** receives first delayed clock **1230** and second input **1236** receives second delayed clock **1232**. Logic of waveform shaping circuit **1222** generates changed clock **1228** exhibiting a rising edge at least partially responsive to first delayed clock **1230** exhibiting a rising edge, and generates changed clock **1228** exhibiting a falling edge at least partially responsive to second delayed clock **1232** exhibiting a falling edge. Waveform shaping circuit **1222** may include, as a non-limiting example, combinational logic circuits to implement generation of changed clock **1228** based on first delayed clock **1230** and second delayed clock **1232** as described herein.

[0117] If mismatch signal **1214** indicates a pulse width of changed clock **1212** is too short, logic circuit **1208** may generate delay setting **1216** to decrement a delay amount at variable delay circuit **1202** and thereby increase a pulse width of changed clock **1228**. If mismatch signal **1214** indicates a pulse width of changed clock **1212** is too long, logic circuit **1208** may generate delay setting **1216** to increment a delay amount at variable delay circuit **1202** and thereby shorten a pulse width of changed clock **1228**.

[0118] FIG. **13A** and FIG. **13B** re timing diagrams respectively depicting signals **1300a** and signals **1300b** during respective example operations of apparatus **1200**.

[0119] Specifically, FIG. **13A** depicts signals including: second delayed clock **1232**, first delayed clock **1230** and changed clock **1212** in an example where a pulse width of original clock **1210** is

shorter than a pulse width of target clock **1224**. FIG. 13B depicts signals including second delayed clock **1232**, first delayed clock **1230** and changed clock **1212** in an example where a pulse width of original clock **1210** is longer than a desired pulse width of target clock **1224**.

[0120] Turning to an example widening of a short pulse depicted by FIG. 13A:

[0121] At time T.sub.1, first delayed clock **1230** exhibits a change from a first state to a second, different, state, here, a rising edge. At time T.sub.1, waveform shaping circuits **1222** generates changed clock **1212** exhibiting a change from a first state to a second state at least partially responsive to detecting the change from the first state to the second state of first delayed clock **1230**.

[0122] At time T.sub.2, second delayed clock **1232** exhibits a change from the first state to the second state, here, a rising edge, but this change of second delayed clock **1232** does not affect changed clock **1212** generation by waveform shaping circuit **1222**.

[0123] At time T.sub.3, first delayed clock **1230** exhibits a change from the second state to the first state, here, a falling edge, but this change of second delayed clock **1232** does not affect changed clock **1212** generation by waveform shaping circuit **1222**.

[0124] At time T.sub.4, second delayed clock **1232** exhibits a change from the second state to the first state, here, a falling edge. At time T.sub.2, waveform shaping circuits **1222** generates changed clock **1212** exhibiting a change from the second state to the first state at least partially responsive to detecting the change from the first state to the second state of first delayed clock **1230**. Time duration ΔT in FIG. 13A may substantially equal one or more delay increments implemented by variable delay circuit **1202** discussed above.

[0125] Turning to an example shortening of a long pulse depicted by FIG. 13B:

[0126] At time T.sub.1, first delayed clock **1230** exhibits a change from a first state to a second, different, state, here, a rising edge, but this change of second delayed clock **1232** does not affect changed clock **1212** generation by waveform shaping circuit **1222**.

[0127] At time T.sub.2, first delayed clock **1230** exhibits a change from a first state to a second, different, state, here, a rising edge. At time T.sub.2, waveform shaping circuits **1222** generates changed clock **1212** exhibiting a change from a first state to a second state at least partially responsive to detecting the change from the first state to the second state of first delayed clock **1230**.

[0128] At time T.sub.3, second delayed clock **1232** exhibits a change from the second state to the first state, here, a falling edge. At time T.sub.3, waveform shaping circuits **1222** generates changed clock **1212** exhibiting a change from the second state to the first state at least partially responsive to detecting the change from the first state to the second state of first delayed clock **1230**. Time duration ΔT in FIG. 13A may substantially equal one or more delay increments implemented by variable delay circuit **1202**, discussed above.

[0129] At time T.sub.4, first delayed clock **1230** exhibits a change from the second state to the first state, here, a falling edge, but this change of first delayed clock **1230** does not affect changed clock **1212** generation by waveform shaping circuit **1222**.

[0130] FIG. 14 is a flow diagram depicting a process **1400** to generate delay settings **1216** to set the amount of delay variable delay circuit **1202** or fixed delay circuit **1204** of FIG. 12, in accordance with one or more examples.

[0131] Although the example process **1400** depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of the process **1400**. In other examples, different components of an example device or system that implements the process **1400** may perform functions at substantially the same time or in a specific sequence.

[0132] At operation **1402**, process **1400** sets respective delay amount at variable delay circuit and fixed delay circuit to a about $\frac{1}{2}$ delay range of the variable delay circuit.

[0133] At operation **1404**, process **1400**, if a desired pulse width is longer than a pulse width of the one of the reference clock or the output clock, generates a delay setting decrementing the delay of the variable delay circuit.

[0134] At operation **1406**, process **1400**, if a desired pulse width is shorter than a pulse width of the one of the reference clock or the output clock, generates a delay setting incrementing the delay of the variable delay circuit.

[0135] FIG. **15** is a flow diagram depicting a process **1500** to generate a change clock via waveform shaping circuit **1222** of FIG. **12**, in accordance with one or more examples.

[0136] Although the example process **1500** depicts a particular sequence of operations, the sequence may be altered without departing from the scope of the present disclosure. For example, some of the operations depicted may be performed in parallel or in a different sequence that does not materially affect the function of the process **1500**. In other examples, different components of an example device or system that implements the process **1500** may perform functions at substantially the same time or in a specific sequence.

[0137] At operation **1502**, process **1500** receives a first delayed clock (e.g., first delayed clock **1230**, without limitation) from the variable delay circuit (e.g., variable delay circuit **1202**, without limitation).

[0138] At operation **1504**, process **1500** receives a second delay clock (e.g., second delayed clock **1232**, without limitation) via the fixed delay circuit (e.g., fixed delay circuit **1204**, without limitation).

[0139] At operation **1506**, process **1500** generates a changed clock (e.g., changed clock **1228**, without limitation) exhibiting a change from a first state to a second state (a rising edge) at least partially responsive to the first delayed clock (e.g., first delayed clock **1230**, without limitation) exhibiting a change from the first state to the second state (a rising edge).

[0140] At operation **1508**, process **1500** generates the changed clock (e.g., changed clock **1228**, without limitation) exhibiting the second state.

[0141] At operation **1510**, process **1500** generates the changed clock (e.g., changed clock **1228**, without limitation) exhibiting a change from the second state to the first state (a falling edge) at least partially responsive to the second delayed clock (e.g., second delayed clocks **1232**, without limitation) exhibiting a change from the second state to the first state (a falling edge).

[0142] It will be appreciated by those of ordinary skill in the art that functional elements of examples disclosed herein (e.g., functions, operations, acts, processes, or methods) may be implemented in any suitable hardware, software, firmware, or combinations thereof. FIG. **16** illustrates non-limiting examples of implementations of functional elements disclosed herein. In some examples, some or all portions of the functional elements disclosed herein may be performed by hardware specially configured for carrying out the functional elements.

[0143] FIG. **16** is a block diagram of a circuit **1600** that, in some examples, may be used to implement various functions, operations, acts, processes, or methods disclosed herein. The circuit **1600** includes one or more processors **1602** (sometimes referred to herein as “processors **1602**”) operably coupled to one or more data storage devices **1604** (sometimes referred to herein as “storage **1604**”). The storage **1604** includes machine-executable code **1606** stored thereon and the processors **1602** include logic circuit **1608**. The machine-executable code **1606** information describing functional elements that may be implemented by (e.g., performed by) the logic circuit **1608**. The logic circuit **1608** is adapted to implement (e.g., perform) the functional elements described by the machine-executable code **1606**. The circuit **1600**, when executing the functional elements described by the machine-executable code **1606**, should be considered as special purpose hardware configured for carrying out functional elements disclosed herein. In some examples the processors **1602** may be configured to perform the functional elements described by the machine-executable code **1606** sequentially, concurrently (e.g., on one or more different hardware platforms), or in one or more parallel process streams.

[0144] When implemented by logic circuit **1608** of the processors **1602**, the machine-executable code **1606** adapts the processors **1602** to perform operations of examples disclosed herein. By way of non-limiting example, the machine-executable code **1606** may adapt the processors **1602** to perform some or a totality of operations of one or more of: process **800**, process **900**, process **1000**, process **1100**, process **1400**, or process **1500**.

[0145] Also by way of non-limiting example, the machine-executable code **1606** may adapt the processors **1602** to perform some or a totality of features, functions, or operations disclosed herein for one or more of: apparatus **100**, including error detector **102**, controller **104**, controlled-oscillator **106**, or duty cycle matching circuit **108**; apparatus **200**, including, comparison circuit **202**, logic circuit **208**, signal path **210**, or duty cycle matching circuit **206**; apparatus **300**, including, variable delay circuit **302**, fixed delay circuit **304**, comparison circuit **306**, logic circuit **308**, waveform shaping circuit **338**, wide logic circuit **320**, skinny logic circuit **322**, or multiplexer **310**. apparatus **600**, including, first filter **602**, second filter **604**, or comparator circuit **606**; apparatus **700**, including first filter **702**, second filter **704**, differential amplifier **706**, duty cycle matching circuit **718**; or apparatus **1200**, including comparison circuit **1206**, logic circuit **1208**, variable delay circuit **1202**, fixed delay circuit **1204**, or waveform shaping circuit **1222**.

[0146] The processors **1602** may include a general purpose processor, a special purpose processor, a central processing unit (CPU), a microcontroller, a programmable logic controller (PLC), a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, other programmable device, or any combination thereof designed to perform the functions disclosed herein. A general-purpose computer including a processor is considered a special-purpose computer while the general-purpose computer is configured to execute functional elements corresponding to the machine-executable code **1606** (e.g., software code, firmware code, hardware descriptions) related to examples of the present disclosure. It is noted that a general-purpose processor (may also be referred to herein as a host processor or simply a host) may be a microprocessor, but in the alternative, the processors **1602** may include any conventional processor, controller, microcontroller, or state machine. The processors **1602** may also be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0147] In some examples the storage **1604** includes volatile data storage (e.g., random-access memory (RAM)), non-volatile data storage (e.g., Flash memory, a hard disc drive, a solid state drive, erasable programmable read-only memory (EPROM), without limitation). In some examples the processors **1602** and the storage **1604** may be implemented into a single device (e.g., a semiconductor device product, a system on chip (SOC), without limitation). In some examples the processors **1602** and the storage **1604** may be implemented into separate devices.

[0148] In some examples the machine-executable code **1606** may include computer-readable instructions (e.g., software code, firmware code). By way of non-limiting example, the computer-readable instructions may be stored by the storage **1604**, accessed directly by the processors **1602**, and executed by the processors **1602** using at least the logic circuit **1608**. Also by way of non-limiting example, the computer-readable instructions may be stored on the storage **1604**, transferred to a memory device (not shown) for execution, and executed by the processors **1602** using at least the logic circuit **1608**. Accordingly, in some examples the logic circuit **1608** includes electrically configurable logic circuit **1608**.

[0149] In some examples the machine-executable code **1606** may describe hardware (e.g., circuitry) to be implemented in the logic circuit **1608** to perform the functional elements. This hardware may be described at any of a variety of levels of abstraction, from low-level transistor layouts to high-level description languages. At a high-level of abstraction, a hardware description language (HDL) such as an IEEE Standard hardware description language (HDL) may be used. By

way of non-limiting examples, Verilog, System Verilog or very large scale integration (VLSI) hardware description language (VHDL) may be used.

[0150] HDL descriptions may be converted into descriptions at any of numerous other levels of abstraction as desired. As a non-limiting example, a high-level description can be converted to a logic-level description such as a register-transfer language (RTL), a gate-level (GL) description, a layout-level description, or a mask-level description. As a non-limiting example, micro-operations to be performed by hardware logic circuits (e.g., gates, flip-flops, registers, without limitation) of the logic circuit **1608** may be described in a RTL and then converted by a synthesis tool into a GL description, and the GL description may be converted by a placement and routing tool into a layout-level description that corresponds to a physical layout of an integrated circuit of a programmable logic device, discrete gate or transistor logic, discrete hardware components, or combinations thereof. Accordingly, in some examples the machine-executable code **1606** may include an HDL, an RTL, a GL description, a mask level description, other hardware description, or any combination thereof.

[0151] In examples where the machine-executable code **1606** includes a hardware description (at any level of abstraction), a system (not shown, but including the storage **1604**) may be configured to implement the hardware description described by the machine-executable code **1606**. By way of non-limiting example, the processors **1602** may include a programmable logic device (e.g., an FPGA or a PLC) and the logic circuit **1608** may be electrically controlled to implement circuitry corresponding to the hardware description into the logic circuit **1608**. Also by way of non-limiting example, the logic circuit **1608** may include hard-wired logic manufactured by a manufacturing system (not shown, but including the storage **1604**) according to the hardware description of the machine-executable code **1606**.

[0152] Regardless of whether the machine-executable code **1606** includes computer-readable instructions or a hardware description, the logic circuit **1608** is adapted to perform the functional elements described by the machine-executable code **1606** when implementing the functional elements of the machine-executable code **1606**. It is noted that although a hardware description may not directly describe functional elements, a hardware description indirectly describes functional elements that the hardware elements described by the hardware description are capable of performing.

[0153] As used in the present disclosure, the terms “module” or “component” may refer to specific hardware implementations configured to perform the actions of the module or component and/or software objects or software routines that may be stored on and/or executed by general purpose hardware (e.g., computer-readable media, processing devices, without limitation) of the computing system. In some examples, the different components, modules, engines, and services described in the present disclosure may be implemented as objects or processes that execute on the computing system (e.g., as separate threads). While some of the system and methods described in the present disclosure are generally described as being implemented in software (stored on and/or executed by general purpose hardware), specific hardware implementations or a combination of software and specific hardware implementations are also possible and contemplated.

[0154] As used in the present disclosure, the term “combination” with reference to a plurality of elements may include a combination of all the elements or any of various different subcombinations of some of the elements. For example, the phrase “A, B, C, D, or combinations thereof” may refer to any one of A, B, C, or D; the combination of each of A, B, C, and D; and any subcombination of A, B, C, or D such as A, B, and C; A, B, and D; A, C, and D; B, C, and D; A and B; A and C; A and D; B and C; B and D; or C and D.

[0155] Terms used in the present disclosure and especially in the appended claims (e.g., bodies of the appended claims, without limitation) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including, but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes, but is not

limited to,” without limitation). As used herein, the term “each” means “some or a totality.” As used herein, the term “each and every” means a “totality.”

[0156] Additionally, if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to examples containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should be interpreted to mean “at least one” or “one or more,” without limitation); the same holds true for the use of definite articles used to introduce claim recitations.

[0157] In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, means at least two recitations, or two or more recitations, without limitation). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, without limitation” or “one or more of A, B, and C, without limitation” is used, in general such a construction is intended to include A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B, and C together, without limitation.

[0158] Further, any disjunctive word or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” should be understood to include the possibilities of “A” or “B” or “A and B.”

[0159] Example 1: A method, comprising: comparing a first value and a second value, the first value representing a duty cycle of a reference clock and the second value representing a duty cycle of an output clock generated by a clock tracking circuit to track the reference clock; setting a duty cycle of a changed clock to reduce duty cycle mismatch between the reference clock and the output clock indicated by the comparing; and providing the changed clock having set duty cycle in lieu of the one of the reference clock or the output clock.

[0160] Example 2: The method according to Example 1, wherein the setting the duty cycle of the changed clock at least partially responsive to the comparing comprises: incrementing or decrementing a delay at a variable delay circuit at least partially responsive to the comparing; delaying one of the reference clock or the output clock at the variable delay circuit to generate a first delayed clock; delaying one of the reference clock or the output clock at a fixed delay circuit to generate a second delayed clock; and shaping a waveform of the changed clock at least partially responsive to the first delayed clock and the second delayed clock.

[0161] Example 3: The method according to any of Examples 1 and 2, comprising: generating a delay setting for the variable delay circuit, the delay setting indicating an increment or a decrement.

[0162] Example 4: The method according to any of Examples 1 through 3, wherein an amount of delay by which the delay at the variable delay circuit is incremented or decremented is a predetermined amount of delay.

[0163] Example 5: The method according to any of Examples 1 through 4, wherein shaping the waveform of the changed clock at least partially responsive to the first delayed clock and the second delayed clock comprises: if a desired pulse width is longer than a pulse width of the one of the reference clock or the output clock, generating a waveform shaping setting indicating the changed clock having a lengthened pulse width.

[0164] Example 6: The method according to any of Examples 1 through 5, wherein shaping the waveform of the changed clock at least partially responsive to the first delayed clock and the

second delayed clock comprises: if a desired pulse width is shorter than a pulse width of the one of the reference clock or the output clock, generating a waveform shaping setting indicating the changed clock having a shortened pulse width.

[0165] Example 7: The method according to any of Examples 1 through 6, comprising: filtering the reference clock to obtain a first direct current (DC) signal, the first DC signal having a value proportional to a pulse width of the reference clock; and filtering the output clock to obtain a second DC signal, the second DC signal proportional to a pulse width of the output clock, wherein the first value comprising the value of the first DC signal and the second value comprising the value of the second DC signal.

[0166] Example 8: An apparatus, comprising: a comparison circuit to receive a reference clock and an output clock generated by a clock tracking circuit to track the reference clock; a duty cycle matching circuit arranged in a signal path of one of the reference clock or the output clock, the matching circuit to set a duty cycle of a changed clock and provide the changed clock in lieu of the one of the reference clock or the output clock; and a logic circuit to set the matching circuit to reduce duty cycle mismatch between the reference clock and the output clock indicated by an output of the comparison circuit.

[0167] Example 9: The apparatus according to Example 8, wherein the matching circuit comprises: a variable delay circuit to generate a first delayed clock at least partially responsive to the one of the reference clock or the output clock; a fixed delay circuit to generate a second delayed clock at least partially responsive to the one of the reference clock or the output clock; and a waveform shaping circuit to shape a waveform of the changed clock at least partially responsive to the first delayed clock and the second delayed clock.

[0168] Example 10: The apparatus according to any of Examples 8 and 9, wherein the logic circuit to set the matching circuit by: generating a delay setting to set a delay of the variable delay circuit; and generating a waveform shaping setting to set a waveform shaping function of the waveform shaping circuit.

[0169] Example 11: The apparatus according to any of Examples 8 through 10, wherein the delay setting indicates an increment or a decrement.

[0170] Example 12: The apparatus according to any of Examples 8 through 11, wherein the waveform shaping circuit comprises: a wide logic circuit to generate a first changed clock having a pulse width shorter than the pulse width of the one of the reference clock or the output clock; a skinny logic circuit to generate a second changed clock having a pulse width longer than the pulse width of the one of the reference clock or the output clock; and a multiplexer to provide one of the first changed clock or the second changed clock, wherein the changed clock comprises the provided one of the first changed clock or the second changed clock.

[0171] Example 13: The apparatus according to any of Examples 8 through 12, wherein the logic circuit to set the matching circuit by: if a desired pulse width is longer than a pulse width of the one of the reference clock or the output clock, generating a delay setting decrementing the delay of the variable delay circuit; and if a desired pulse width is shorter than a pulse width of the one of the reference clock or the output clock, generate a delay setting incrementing the delay of the variable delay circuit.

[0172] Example 14: The apparatus according to any of Examples 8 through 13, wherein the waveform shaping circuit to: generate the changed clock exhibiting a change from a first state to a second state at least partially responsive to the first delayed clock exhibiting a change from the first state to the second state; and generate the changed clock exhibiting a change from the second state to the first state at least partially responsive to the second delayed clock exhibiting a change from the second state to the first state.

[0173] Example 15: The apparatus according to any of Examples 8 through 14, comprising: a comparison circuit to receive a reference clock and an output clock generated by a clock tracking circuit to track the reference clock; a duty cycle matching circuit arranged in a signal path of one of

the reference clock or the output clock, the matching circuit to set a duty cycle of a changed clock and provide the changed clock in lieu of the one of the reference clock or the output clock; and a logic circuit to set the matching circuit to reduce duty cycle mismatch between the reference clock and the output clock indicated by an output of the comparison circuit.

[0174] Example 16: The apparatus according to any of Examples 8 through 15, wherein the comparison circuit comprises: a first filter to filter the reference clock to obtain a first direct current (DC) signal, the first DC signal proportional to a pulse width of the reference clock; a second filter to filter the output clock to obtain a second DC signal, the second DC signal proportional to a pulse width of the output clock; and a comparator to compare the first DC signal and the second DC signal and generate an indication of mismatch between the first DC signal and the DC signal.

[0175] Example 17: The apparatus according to any of Examples 8 through 16, wherein the clock tracking circuit is chosen from a group, the group including: a hybrid phase locked loop (PLL), an analog PLL, a digital PLL, a delay locked loop, an injection locked loop, or a frequency synthesizer.

[0176] Example 18: An apparatus, comprising: a differential amplifier coupled to receive a reference clock and an output clock generated by a clock tracking circuit to track the reference clock; and a duty cycle matching circuit arranged in a signal path of one of the reference clock or the output clock, the matching circuit to: set a duty cycle of a changed clock at least partially responsive to duty cycle mismatch indicated by an output of the differential amplifier; and provide the changed clock in lieu of the one of the reference clock or the output clock.

[0177] Example 19: The apparatus according to Example 18, wherein the matching circuit comprises: a variable delay circuit having a control input directly coupled to an output of the differential amplifier; a fixed delay circuit; and a waveform shaping circuit having an input directly coupled to an output of the differential amplifier.

[0178] While the present disclosure has been described herein with respect to certain illustrated examples, those of ordinary skill in the art will recognize and appreciate that the present invention is not so limited. Rather, many additions, deletions, and modifications to the illustrated and described examples may be made without departing from the scope of the invention as hereinafter claimed along with their legal equivalents. In addition, features from one example may be combined with features of another example while still being encompassed within the scope of the invention as contemplated by the inventor.

Claims

1. A method, comprising: setting, at least partially based on an indication of mismatch between duty cycles of a first clock and a second clock generated to track the first clock, a duty cycle of a third clock; and providing the third clock having set duty cycle to an error detector of the clock tracking circuit in place of one of the first clock or the second clock.
2. The method of claim 1, wherein the setting the duty cycle of the changed clock at least partially responsive to the comparing comprises: incrementing or decrementing a delay at a variable delay circuit at least partially responsive to the comparing; delaying one of the reference clock or the output clock at the variable delay circuit to generate a first delayed clock; delaying one of the reference clock or the output clock at a fixed delay circuit to generate a second delayed clock; and shaping a waveform of the changed clock at least partially responsive to the first delayed clock and the second delayed clock.
3. The method of claim 2, comprising: generating a delay setting for the variable delay circuit, the delay setting indicating an increment or a decrement.
4. The method of claim 3, wherein an amount of delay by which the delay at the variable delay circuit is incremented or decremented is a predetermined amount of delay.
5. The method of claim 2, wherein shaping the waveform of the changed clock at least partially

responsive to the first delayed clock and the second delayed clock comprises: if a desired pulse width is longer than a pulse width of the one of the reference clock or the output clock, generating a waveform shaping setting indicating the changed clock having a lengthened pulse width.

6. The method of claim 2, wherein shaping the waveform of the changed clock at least partially responsive to the first delayed clock and the second delayed clock comprises: if a desired pulse width is shorter than a pulse width of the one of the reference clock or the output clock, generating a waveform shaping setting indicating the changed clock having a shortened pulse width.

7. The method of claim 1, comprising: filtering the reference clock to obtain a first direct current (DC) signal, the first DC signal having a value proportional to a pulse width of the reference clock; and filtering the output clock to obtain a second DC signal, the second DC signal proportional to a pulse width of the output clock, wherein the first value comprising the value of the first DC signal and the second value comprising the value of the second DC signal.

8. The method of claim 1, further comprising comparing a duty cycle parameter of the first clock with a duty cycle parameter of the second clock to generate the indication of mismatch.

9. An apparatus for reducing duty cycle mismatch among clock signals, comprising: means for setting, at least partially based on an indication of mismatch between duty cycles of a first clock and a second clock generated to track the first clock, a duty cycle of a third clock; and means for providing the third clock having the set duty cycle to an error detector of the clock tracking circuit in place of at least one of the first clock or the second clock.

10. The apparatus of claim 9, wherein the matching circuit comprises: a variable delay circuit to generate a first delayed clock at least partially responsive to the one of the reference clock or the output clock; a fixed delay circuit to generate a second delayed clock at least partially responsive to the one of the reference clock or the output clock; and a waveform shaping circuit to shape a waveform of the changed clock at least partially responsive to the first delayed clock and the second delayed clock.

11. The apparatus of claim 10, wherein the logic circuit to set the matching circuit by: generating a delay setting to set a delay of the variable delay circuit; and generating a waveform shaping setting to set a waveform shaping function of the waveform shaping circuit.

12. The apparatus of claim 11, wherein the delay setting indicates an increment or a decrement.

13. The apparatus of claim 10, wherein the waveform shaping circuit comprises: a wide logic circuit to generate a first changed clock having a pulse width shorter than the pulse width of the one of the reference clock or the output clock; a skinny logic circuit to generate a second changed clock having a pulse width longer than the pulse width of the one of the reference clock or the output clock; and a multiplexer to provide one of the first changed clock or the second changed clock, wherein the changed clock comprises the provided one of the first changed clock or the second changed clock.

14. The apparatus of claim 10, wherein the logic circuit to set the matching circuit by: if a desired pulse width is longer than a pulse width of the one of the reference clock or the output clock, generating a delay setting decrementing the delay of the variable delay circuit; and if a desired pulse width is shorter than a pulse width of the one of the reference clock or the output clock, generate a delay setting incrementing the delay of the variable delay circuit.

15. The apparatus of claim 14, wherein the waveform shaping circuit to: generate the changed clock exhibiting a change from a first state to a second state at least partially responsive to the first delayed clock exhibiting a change from the first state to the second state; and generate the changed clock exhibiting a change from the second state to the first state at least partially responsive to the second delayed clock exhibiting a change from the second state to the first state.

16. The apparatus of claim 10, comprising: a comparison circuit to receive a reference clock and an output clock generated by a clock tracking circuit to track the reference clock; a duty cycle matching circuit arranged in a signal path of one of the reference clock or the output clock, the matching circuit to set a duty cycle of a changed clock and provide the changed clock in lieu of the

one of the reference clock or the output clock; and a logic circuit to set the matching circuit to reduce duty cycle mismatch between the reference clock and the output clock indicated by an output of the comparison circuit.

17. The apparatus of claim 9, wherein the comparison circuit comprises: a first filter to filter the reference clock to obtain a first direct current (DC) signal, the first DC signal proportional to a pulse width of the reference clock; a second filter to filter the output clock to obtain a second DC signal, the second DC signal proportional to a pulse width of the output clock; and a comparator to compare the first DC signal and the second DC signal and generate an indication of mismatch between the first DC signal and the DC signal.

18. The apparatus of claim 9, wherein the clock tracking circuit is chosen from a group, the group including: a hybrid phase locked loop (PLL), an analog PLL, a digital PLL, a delay locked loop, an injection locked loop, or a frequency synthesizer.

19. The apparatus of claim 9, further comprising a comparator to compare a duty cycle parameter of the first clock with a duty cycle parameter of the second clock to generate the indication of mismatch.
